

Clinical Research Report

PATIENT & PHYSICIAN

Patient Information

Last Name: LAST
First Name: FIRST
Sex At Birth: SEX
Date of Birth: YYYY-MM-DD

Physician Information

Name: LAST, FIRST
Hospital: HOSPITAL NAME
Licence #: nnnnnnnn
Phone #: nnn-nnn-nnnn
Requisitioner Email: NAME@domain.com

CASE OVERVIEW

Assay: Targeted Sequencing - REVOLVE Panel - cfDNA and Buffy Coat (v5.0)

Primary cancer: None

Site of biopsy/surgery: cfDNA

Study: PLACEHOLDER Patient Study ID: PLACEHOLDER

Patient LIMS ID: PLACEHOLDER Plasma Sample ID: PLACEHOLDER

Requisition Approved: 2026-01-01 Blood Sample ID: PLACEHOLDER

Date Report Prepared: 2026-01-01 Report ID: PLACEHOLDER-v1

TREATMENT OPTIONS

Review identified 0 option(s) indicating an FDA-approved and/or NCCN-compendium listed treatment, 0 option(s) indicating investigational therapies, and 0 option(s) indicating NCCN-listed biomarkers.

CTDNA STATUS

Cell-free DNA tumour burden: Not Detected

Copy number analysis: ctDNA not detected

Small mutation analysis: ctDNA not detected

RESULTS SUMMARY

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SAMPLE INFORMATION

OncoTree code:	N/A	Known Variants:	PLACEHOLDER
Sample Type:	cfDNA	Estimated Cancer Cell Content (%):	<10
Raw Coverage (mean):	16,857	Unique Molecular Coverage (mean):	2,919

SNVS AND IN/DELS

No somatic oncogenic mutations were uncovered.

SHALLOW WHOLE GENOME SEQUENCING

The estimated coverage-derived tumour fraction was below the limit of 10% for reporting copy number variants.

ASSAY
DESCRIPTION

This assay combines two comprehensive next generation sequencing assays: a targeted sequencing (TAR) assay and a shallow whole genome sequencing (sWGS) assay. Shallow whole genome sequencing (sWGS) libraries were prepared using the KAPA Hyper Prep kit with DNA extracted from FFPE, cfDNA, fresh frozen tissue (for tumour samples). Paired-end sequencing was performed using the NextSeq 2000 illumina technology to a minimum target coverage of 0.1x. Alignments were performed using [bwamem2](#) (2.2.1) against reference genome [GRCh38.p12](#). Copy number amplifications were called using [ichorCNA](#). TAR libraries were prepared using the KAPA Hyper Prep kit with DNA extracted from FFPE, cfDNA, fresh frozen tissue (for tumour samples) or buffy coat blood specimens (for matched normal blood samples). Paired-end sequencing was performed using the NextSeq 2000 illumina technology. Alignments to the reference genome [GRCh38.p12](#) with [bwamem2](#) (2.2.1) followed by unique molecular based error suppression with [ConsensusCruncher](#) were performed through our umiConsensus workflow. SNVs and INDELs were called using [MuTect2](#) (GATK 4.1.6.0) through our mutect2consensus workflow. Fourteen genes were annotated with [VariantEffectPredictor](#) (105.0) using transcripts from [MANE Select](#) version 1.0. Variants were further annotated for oncogenicity and actionability by [OncoKB](#). Variants without an OncoKB actionable tier were reported if they were listed as oncogenic, likely oncogenic or predicted oncogenic. Assay results were collated into the report document by [Djerba](#) (1.13.0) using TAR pipeline 5.0.

TARGET
GROUP
INFORMATION

Target Group	Probes	Space (bp)	Gene	Transcript	Span
Hereditary Breast and Ovarian Cancer (HBOC)	937	112,440	BRCA2	NM_000059.4	all exons and introns
			BRCA1	NM_007294.3	all exons and introns
			PALB2	NM_024675.3	all exons
			TP53	NM_000546.5	all exons
Lynch Syndrome	274	32,880	APC	NM_000038.5	all exons
			EPCAM	NM_002354.2	exons and 3'-UTR
			PMS2	NM_000535.6	all exons
			MLH1	NM_000249.3	all exons
			MSH2	NM_000251.2	all exons
			MSH6	NM_000179.2	all exons
			CCNE1	NM_001238.4	all exons
Hereditary Cancer Panel Expansion	148	17,760	CDH1	NM_004360.5	all exons
			VHL	NM_000551.4	all exons
			NF1	NM_001042492.3	all exons

DISCLAIMER

For calling amplifications from sWGS, the sensitivity is 74.2% and the specificity is 99.4% when recalling amplifications known from prior tissue-based sWGS. The limit of detection is ≥ 1.4 -fold change for gene amplifications in a tumour with 10% cancer cell content. For SNV/INDEL detection in FFPE tumours, the sensitivity is 95.5% and the specificity is 94.1%. For SNV/INDEL detection in cfDNA samples, the sensitivity is 84% and the specificity is 97%. The limit of detection was determined to be 1% at a minimum collapsed coverage of 400x, with at least 3 supporting reads. Laboratory results are subject to approximately 0.5% error in any of the pre-analytical, analytical or post-analytical phases of the test ([Bonini et al. 2002](#)). This test was developed and its performance characteristics determined by OICR Genomics. It has not been cleared or approved by the US Food and Drug Administration.

REPORT
SIGN-OFFS

Report drafted by CGI Reporter on 2026-01-01
Report electronically signed out by PLACEHOLDER (ABMS #XXXXXXX) on yyyy-mm-dd

APPENDIX

Actionability Definitions

Variant prioritization is based on OncoKB actionability tiers:

Level	Definition
1	FDA-recognized biomarker predictive of response to an FDA-approved drug in this indication
2	Standard care biomarker recommended by the NCCN or other expert panels predictive of response to an FDA-approved drug in this indication
3A	Compelling clinical evidence supports the biomarker as being predictive of response to a drug in this indication
3B	Standard care or investigational biomarker predictive of response to an FDA-approved or investigational drug in another indication
4	Compelling biological evidence supports the biomarker as being predictive of response to a drug
R1	Standard care biomarker predictive of resistance to an FDA-approved drug in this indication
R2	Compelling clinical evidence supports the biomarker as being predictive of resistance to a drug
N1	The biomarker is listed as "Oncogenic" by OncoKB, but is not in an actionable tier
N2	The biomarker is listed as "Likely Oncogenic" by OncoKB, but is not in an actionable tier
N3	The biomarker is listed as "Predicted Oncogenic" by OncoKB, but is not in an actionable tier
N4	The biomarker is listed as "Likely Neutral" or "Inconclusive" by OncoKB
P	The biomarker is listed as "Prognostic" by OncoKB or NCCN

When provided, results and interpretations are consistent with available knowledge for the given tumour type as defined in [OncoTree](#). OncoKB tiers are tumour-specific and dependent on OncoTree definitions.

Definitions

Raw Coverage (Mean): Average number of bases covering each sequenced base. Target coverage is 15,000x for the tumour, and 5,000x for the normal.

Unique Molecular Coverage(Mean): Average number of bases covering each sequenced base after Unique molecule based error suppression. Minimum coverage is 400x for the tumour, and 400x for the normal.

Estimated Cancer Cell Content (%): The inferred tumour purity as determined using ichorCNA

Copy State: Copy state changes were called according to the log₂ coverage-ratio, which can be used to calculate an approximate minimum copy number for a diploid tumour with 100% cancer cell content. As the log₂ coverage-ratio thresholds scale with tumour purity, absolute copy number may be lower than predicted, in tumours with lower cancer cell content. Calls represent levels derived from copy-number analysis, are used for annotation by OncoKB, and indicate the copy-number level per gene, as follows:

Amplification a high-level amplification (more copies, often focal)

Gene Descriptions

Gene	Summary
NA	No reportable genes found